

Flight Delay Prediction System using Machine Learning

Model Evaluation Report

Introduction

Flight delays significantly impact airline operations, airport management, and passenger satisfaction. This report evaluates the performance of Logistic Regression and Random Forest models developed for flight delay prediction.

Objective of Model Evaluation

The primary objective is to assess model performance in predicting whether a flight will be delayed by more than 15 minutes.

Dataset Summary

Dataset Source: Kaggle US Flight Delay Dataset.

Sample Size: 100000 rows

Target Variable: DELAYED = ARRIVAL_DELAY > 15 minutes

Data Preprocessing Overview

Missing value handling Mean and median imputation Label encoding Feature scaling Class balancing Data leakage removal Train-test split

Class Imbalance Handling

Class imbalance was addressed using `class_weight='balanced'`. This improved recall for delayed flights and reduced bias toward the majority class.

Evaluation Metrics

Accuracy: Overall prediction correctness.

Precision: Reliability of predicted delays.

Recall: Ability to detect actual delays.

F1-score: Balance between precision and recall.

ROC-AUC: Overall classification capability.

Confusion Matrix: Displays prediction performance.

Logistic Regression Results

Accuracy: 71.6%
ROC-AUC: 0.79

Precision: Class 0: 0.83 Class 1: 0.56 Recall: Class 0: 0.72 Class 1: 0.71 F1-score: Class 0: 0.77 Class 1: 0.63
Confusion Matrix: [[9574, 3695], [1984, 4747]]

Random Forest Results

Accuracy: 83.23%
ROC-AUC: 0.89

Precision: Class 0: 0.86 Class 1: 0.77 Recall: Class 0: 0.89 Class 1: 0.71 F1-score: Class 0: 0.88 Class 1: 0.74
Confusion Matrix: [[11869, 1400], [1953, 4778]]

ROC Curve Comparison

Random Forest achieved a higher ROC-AUC score of 0.89 compared to Logistic Regression (0.79), indicating superior discrimination capability.

Importance of Recall

Recall is highly important in airline systems because missed delays can negatively impact airport operations, scheduling, and passenger management.

Model Comparison

Metric	Logistic Regression	Random Forest
Accuracy	71.6%	83.23%
Recall	0.71	0.71
F1-score	0.63	0.74
ROC-AUC	0.79	0.89

Business Impact

Improved operational planning Enhanced passenger communication Better gate and crew allocation Reduced operational costs

Future Improvements

XGBoost implementation Weather data integration Real-time prediction systems Deep learning architectures Cloud deployment pipelines

Conclusion

Random Forest outperformed Logistic Regression with stronger accuracy, F1-score, and ROC-AUC performance. The system demonstrates strong potential for airline operational decision support.

Sample Training Code

```
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(
    n_estimators=300,
    class_weight='balanced',
    random_state=42
)

rf_model.fit(X_train, y_train)
```